P2 Chapter 8: Parametric Equations

Angular Parameters

Using Trig Identities

It's often helpful to use $\sin^2 t + \cos^2 t \equiv 1$ or $1 + \tan^2 t \equiv \sec^2 t$ to turn parametric equations into a single Cartesian one.

[Textbook] A curve has the parametric sequences $x = \sin t + 2$, $y = \cos t - 3$, $t \in \mathbb{R}$.

a) Find a Cartesian equation for the curve.

b) Hence sketch the curve.

?

[Textbook] A curve is defined by the parametric equations

$$x = \sin t\,, \qquad y = \sin 2t\,, \qquad -\frac{\pi}{2} \le t \le \frac{\pi}{2}$$

a) Find a Cartesian equation of the curve in the form $y = f(x)$, $-k \le x \le k$, stating the value of the constant k.

b) Write down the range of $f(x)$.

a

?

b

?

Using Trig Identities

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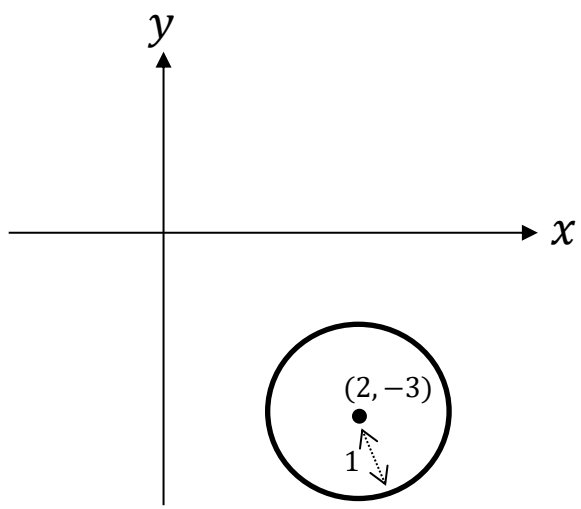
a) Find a Cartesian equation for the curve.

b) Hence sketch the curve.

$$\sin^2 t + \cos^2 t \equiv 1$$
$$\therefore (x-2)^2 + (y+3)^2 = 1$$



[Textbook] A curve is defined by the parametric equations

$$x = \sin t, \qquad y = \sin 2t, \qquad -\frac{\pi}{2} \le t \le \frac{\pi}{2}$$

a) Find a Cartesian equation of the curve in the form $y = f(x)$, $-k \le x \le k$, stating the value of the constant k.

b) Write down the range of $f(x)$.

a $y = 2\sin t \cos t$

$= 2\sin t\sqrt{1-\sin^2 t}$

$= 2x\sqrt{1-x^2}$

If we have $\cos t$, we need to turn in into $\sin t$ so that we can replace it with x. It's quite common to have to interchange between $\sin t$ and $\cos t$ in this way. You may find it helpful to write out the identity $\sin^2 t + \cos^2 t = 1$ first.

When $t = -\frac{\pi}{2}$,

$x = \sin\left(-\frac{\pi}{2}\right) = -1.$

When $t = \frac{\pi}{2}, x = \sin\left(\frac{\pi}{2}\right) = 1$

$\therefore k = 1$

b $\sin(2t)$ varies between -1 and 1 when $-\frac{\pi}{2} \le t \le \frac{\pi}{2}$, therefore range is $-1 \le y \le 1$.

Test Your Understanding

C4 June 2013

which double angle formula would be best here?

4. A curve C has parametric equations

$$x = 2\sin t, \qquad y = 1 - \cos 2t, \qquad -\frac{\pi}{2} \le t \le \frac{\pi}{2}$$

(*b*) Find a cartesian equation for C in the form

$$y = f(x), \qquad -k \le x \le k,$$

stating the value of the constant k.

?

[Textbook] A curve C has parametric equations

$$x = \cot t + 2, \qquad y = cosec^2\, t - 2, \qquad 0 < t < \pi$$

a) Find the equation of the curve in the form $y = f(x)$ and state the domain of x for which the curve is defined.

b) Hence, sketch the curve.

? a

? b

Test Your Understanding

C4 June 2013

which double angle formula would be best here?

4. A curve C has parametric equations

$$x = 2\sin t, \qquad y = 1 - \cos 2t, \qquad -\frac{\pi}{2} \le t \le \frac{\pi}{2}$$

(*b*) Find a cartesian equation for C in the form

$$y = f(x), \qquad -k \le x \le k,$$

stating the value of the constant k.

$\sin t = \frac{x}{2}$

$y = 1 - (1 - 2\sin^2 t)$

$= 2\sin^2 t$

$= 2\left(\frac{x}{2}\right)^2$

$= \frac{x^2}{2}$

$k = 2$

[Textbook] A curve C has parametric equations

$$x = \cot t + 2, \qquad y = cosec^2\, t - 2, \qquad 0 < t < \pi$$

a) Find the equation of the curve in the form $y = f(x)$ and state the domain of x for which the curve is defined.

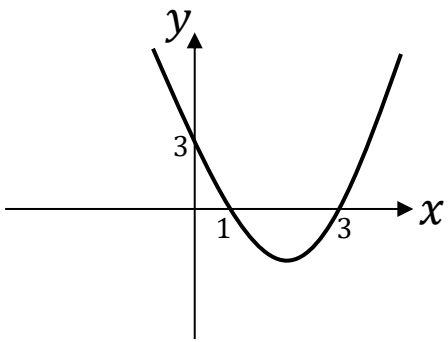
b) Hence, sketch the curve.

$1 + \cot^2 t \equiv cosec^2\, t$

$\therefore 1 + (x-2)^2 = y + 2$

$y = x^2 - 4x + 3$

As t varies between 0 and π, $x = \cot t + 2$ varies over all real numbers (sketch cot to see this). Therefore the domain of $f(x)$ is $x \in \mathbb{R}$.



Exercise 8.2

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Further Exam Practice

C4 June 2012 Q6

Figure 2 shows a sketch of the curve C with parametric equations

$$x = \sqrt{3}\sin 2t, \qquad y = 4\cos^2 t, \qquad 0 \le t \le \pi.$$

(c) Find a cartesian equation of C.

(3)

?

Exercise 8.2

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Further Exam Practice

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Figure 2 shows a sketch of the curve C with parametric equations

$$x = \sqrt{3}\sin 2t, \qquad y = 4\cos^2 t, \qquad 0 \le t \le \pi.$$

(c) Find a cartesian equation of C.

(3)

$$x = 2\sqrt{3}\sin t\cos t$$
$$x^2 = 12\sin^2 t\cos^2 t$$
$$x^2 = 12(1-\cos^2 t)\cos^2 t$$
$$x^2 = 12\left(1-\frac{y}{4}\right)\left(\frac{y}{4}\right)$$

Homework Exercise

1 Find the Cartesian equation of the curves given by the following parametric equations:

a $x = 2\sin t - 1, \quad y = 5\cos t + 4, \quad 0 < t < 2\pi$

b $x = \cos t, \quad y = \sin 2t, \quad 0 < t < 2\pi$

c $x = \cos t, \quad y = 2\cos 2t, \quad 0 < t < 2\pi$

d $x = \sin t, \quad y = \tan 2t, \quad 0 < t < \frac{\pi}{2}$

e $x = \cos t + 2, \quad y = 4\sec t, \quad 0 < t < \frac{\pi}{2}$

f $x = 3\cot t, \quad y = \operatorname{cosec} t, \quad 0 < t < \pi$

2 A circle has parametric equations $x = \sin t - 5, \quad y = \cos t + 2$

a Find a Cartesian equation of the circle.

b Write down the radius and the coordinates of the centre of the circle.

c Write down a suitable domain of t which defines one full revolution around the circle.

Problem-solving

Think about how x and y change as t varies.

3 A circle has parametric equations $x = 4\sin t + 3, \quad y = 4\cos t - 1$. Find the radius and the coordinates of the centre of the circle.

4 A curve is given by the parametric equation $x = \cos t - 2, \quad y = \sin t + 3, \quad -\pi < t < \pi$. Sketch the curve.

Homework Exercise

5 Find the Cartesian equation of the curves given by the following parametric equations.

a $x = \sin t, \quad y = \sin\left(t + \frac{\pi}{4}\right), \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$

b $x = 3\cos t, \quad y = 2\cos\left(t + \frac{\pi}{6}\right), \quad 0 < t < \frac{\pi}{3}$

c $x = \sin t, \quad y = 3\sin(t + \pi), \quad 0 < t < 2\pi$

Hint Use the addition formulae and exact values.

6 The curve C has parametric equations

$$x = 8\cos t, \quad y = \frac{1}{4}\sec^2 t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

a Find a Cartesian equation of C. **(4 marks)**

b Sketch the curve C on the appropriate domain. **(3 marks)**

7 A curve has parametric equations

$$x = 3\cot^2 2t, \quad y = 3\sin^2 2t, \quad 0 < t \leqslant \frac{\pi}{4}$$

Find a Cartesian equation of the curve in the form $y = \mathrm{f}(x)$. State the domain on which $\mathrm{f}(x)$ is defined. **(6 marks)**

8 A curve C has parametric equations

$$x = \frac{1}{3}\sin t, \quad y = \sin 3t, \quad 0 < t < \frac{\pi}{2}$$

a Show that the Cartesian equation of the curve is given by

$$y = ax(1 - bx^2)$$

where a and b are integers to be found. **(5 marks)**

b State the domain and range of $y = \mathrm{f}(x)$ in the given domain of t. **(2 marks)**

Homework Exercise

9 Show that the curve with parametric equations

$$x = 2\cos t, \quad y = \sin\left(t - \frac{\pi}{6}\right), \quad 0 < t < \pi$$

can be written in the form

$$y = \frac{1}{4}\left(\sqrt{12 - 3x^2} - x\right), -2 < x < 2$$ **(6 marks)**

10 A curve has parametric equations

$$x = \tan^2 t + 5, \quad y = 5\sin t, \quad 0 < t < \frac{\pi}{2}$$

a Find the Cartesian equation of the curve in the form $y^2 = f(x)$. **(4 marks)**

b Determine the possible values of x and y in the given domain of t. **(2 marks)**

11 A curve C has parametric equations

$$x = \tan t, \quad y = 3\sin(t - \pi), \quad 0 < t < \frac{\pi}{2}$$

Find a Cartesian equation of C. **(4 marks)**

Challenge

The curve C is given by the parametric equations:

$$x = \frac{1}{2}\cos 2t, \quad y = \sin\left(t + \frac{\pi}{6}\right), \quad 0 < t < 2\pi$$

Show that a Cartesian equation for C is $(4y^2 - 2 + 2x)^2 + 12x^2 - 3 = 0$.

Homework Answers

1 **a** $25(x+1)^2 + 4(y-4)^2 = 100$

b $y^2 = 4x^2(1-x^2)$

c $y = 4x^2 - 2$

d $y = \dfrac{2x\sqrt{1-x^2}}{1-2x^2}$

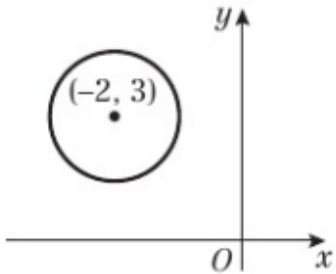
e $y = \dfrac{4}{x-2}$

f $y^2 = 1 + \left(\dfrac{x}{3}\right)^2$

2 **a** $(x+5)^2 + (y-2)^2 = 1$
b Centre $(-5, 2)$, radius 1
c $0 \leqslant t < 2\pi$

3 Centre $(3, -1)$, radius 4

4 $(x+2)^2 + (y-3)^2 = 1$



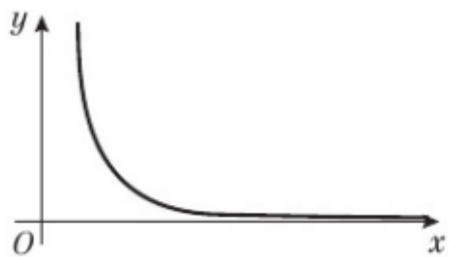
5 **a** $y = \dfrac{\sqrt{2}}{2}x + \dfrac{\sqrt{2(1-x^2)}}{2}$

b $y = \dfrac{\sqrt{3}}{3}x - \dfrac{\sqrt{9-x^2}}{3}, \ \dfrac{3}{2} < x < 3$

c $y = -3x, \ -1 \leqslant x \leqslant 1$

6 **a** $y = \dfrac{16}{x^2}, \ 0 < x \leqslant 8$

b



7 $y = \dfrac{9}{3+x}$ Domain: $x > 0$

8 **a** $y = 9x(1-12x^2) \Rightarrow a = 9, b = 12$
b Domain: $0 < x < \frac{1}{3}$, Range: $-1 < y \leqslant 1$

9 $y = \sin t \cos\left(\dfrac{\pi}{6}\right) - \cos t \sin\left(\dfrac{\pi}{6}\right)$

$$= \frac{\sqrt{3}}{2}\sin t - \frac{1}{2}\cos t = \frac{\sqrt{3\left(1 - \frac{x^2}{4}\right)}}{2} - \frac{1}{4}x$$

$$= \frac{1}{4}\left(2\sqrt{3 - \frac{3}{4}x^2} - x\right) = \frac{1}{4}\left(\sqrt{12 - 3x^2} - x\right)$$

$t = 0 \Rightarrow x = 2, \ t = \pi \Rightarrow x = -2$, so $-2 < x < 2$.

10 **a** $y^2 = 25\left(1 - \dfrac{1}{x-4}\right)$

b $x > 5, \ 0 < y < 5$

11 $x = -\dfrac{y}{\sqrt{9-y^2}}, \ x > 0$

Challenge

$(4y^2 - 2 + 2x)^2 + 12x^2 - 3 = 0$